

ESERCIZI:

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\operatorname{tg} \frac{1}{n^5+5}\right) \sqrt{\operatorname{arctg} \frac{n+2}{n^2-3}} \sinh\left(\frac{n^{5/3}+n}{n^2+n^{13/6}}\right) \cdot \log(\log \sinh n) - 1}{\sqrt[4]{e^{\frac{1}{(-1)^n+n^2}} - 1} \cdot \log(n^5 - n^2)} \quad [= -\frac{6}{5}]$$

$$\lim_{n \rightarrow \infty} \frac{(n! + n^n) \left[(n^2 - \operatorname{arctg} n)^3 - n^2 \right]^{-\frac{1}{8}} - 1}{\log(27n) \left[e^{\frac{1}{n^4}} - \cos\left(\frac{\sin n - n}{[\log(1 + \sinh n)]^2}\right)^2 \right]^{\frac{1}{6}}}} \quad [= \infty]$$

Trovare **d** t.c. il seguente limite sia finito (e calcolarlo)

$$\lim_{n \rightarrow \infty} \frac{(n! + n^n) \left[(n^2 - \operatorname{arctg} n)^3 - n^2 \right]^d - 1}{\log(27n) \left[e^{\frac{1}{n^4}} - \cos\left(\frac{\sin n - n}{[\log(1 + \sinh n)]^2}\right)^2 \right]^{\frac{1}{6}}}} \quad \begin{matrix} [d = -\frac{2}{9}] \\ [= 2\frac{1}{6}] \end{matrix}$$

Trovare **B** t.c. il seguente limite sia finito (e calcolarlo)

$$\lim_{n \rightarrow \infty} \frac{(n! + n^n) \left[(n^2 - \operatorname{arctg} n)^3 - n^2 \right]^{-\frac{1}{8}} - 1}{\log(27n) \left[e^{\frac{1}{n^4}} - \cos\left(\frac{\sin n - n}{[\log(1 + \sinh n)]^2}\right)^2 \right]^{\frac{1}{6}}}} \quad \begin{matrix} [\neq B] \\ [\rightarrow +\infty \text{ V.B.E. } R] \\ B \end{matrix}$$